Computer Science

FARM-IT

CONCEPTUAL DESIGNERS

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Table of Contents

Introduction	3
Softwares Used	4
Flow diagram	5
Source Code	6
Application Flow	32
Home Screens Main window Login window Admin dashboard Customer dashboard Farmer dashboard Register window	33
Report Module Admin screen Customer screen Farmer screen Register screen	37
Files part of this project Python file Exe file	44
Bibliography	45

Introduction

Agriculture forms the backbone of the Indian economy, but farmers often face challenges in marketing their produce and receiving fair prices due to the presence of intermediaries. To address this issue, **Farm-It** has been developed as a Python-based desktop application that enables farmers to directly showcase their products to customers, eliminating middlemen and maximizing their profits.

Farm-It provides a simple, user-friendly interface designed using Python's Tkinter module, making it accessible for both farmers and customers. The application allows farmers to upload their product details such as name, type, price, contact information, and location. Customers can search for products using filters such as product name, tags (e.g., #cereals, #vegetables), price range, and location, enabling them to easily find what they need. Additionally, customers can maintain a personalized wishlist for future reference.

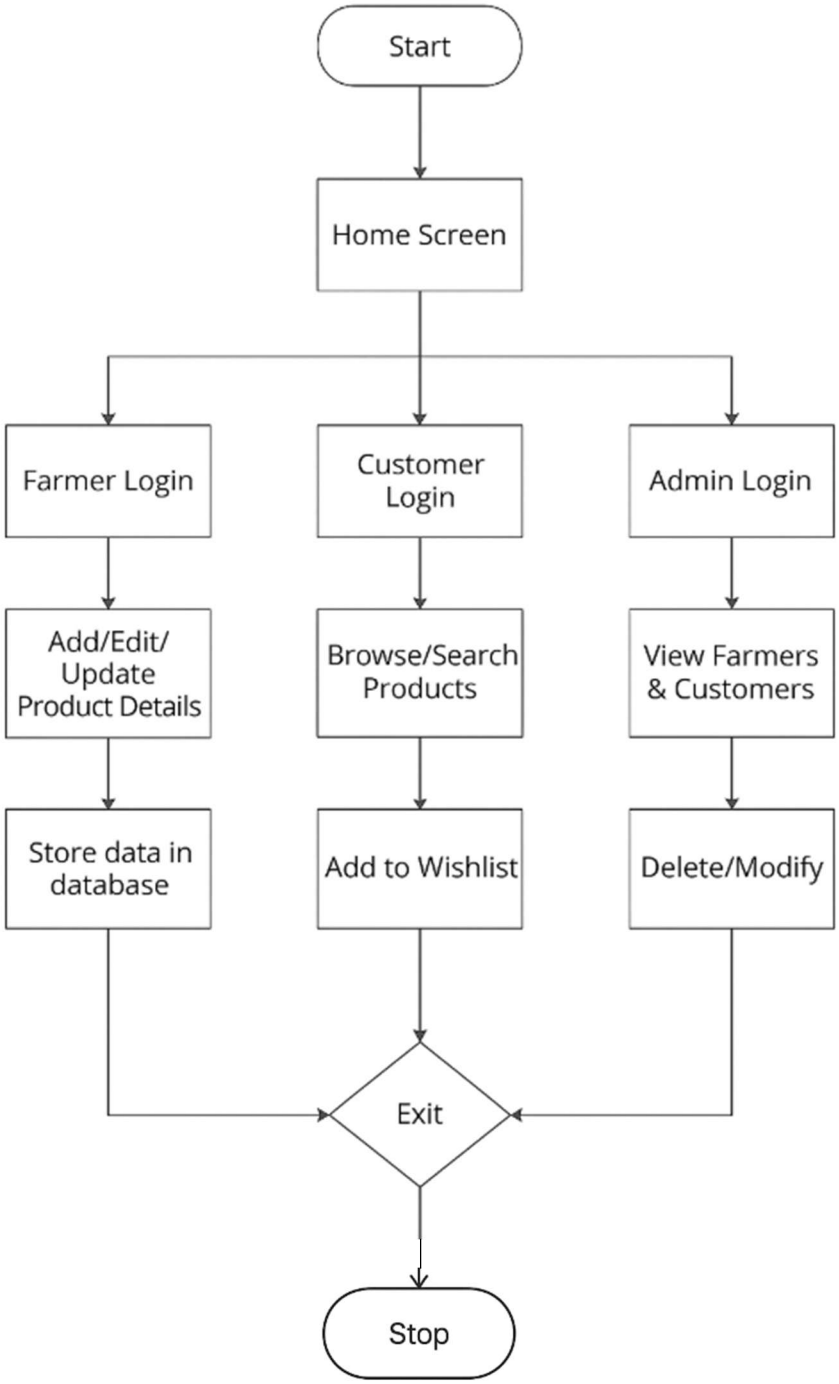
The application also provides an **Admin dashboard** where administrators can monitor all farmer and customer accounts, manage product listings, and delete or update any inappropriate entries. All the data—farmer details, customer details, and product information—is stored in a MySQL database, ensuring security and easy access to information.

Farm-It has been built as a **Grade 12 Computer Science project** to demonstrate the integration of Python programming with database connectivity using MySQL. The project not only simplifies agricultural trading but also showcases how technology can empower farmers and improve the agricultural supply chain.

Softwares Used

1. **Python 3.13**
    - Used as the main programming language for developing the Farm-It application.
    - Python's simplicity, readability, and vast library support made it ideal for building a GUI-based desktop application.
2. **Tkinter**
    - A built-in Python library used for creating the graphical user interface (GUI).
    - Tkinter widgets such as labels, buttons, frames, and entry fields were used to design the farmer, customer, and admin dashboards.
3. **MySQL Database**
    - Used for storing and managing all project data such as farmer profiles, customer details, product listings, and wishlist entries.
    - MySQL Connector/Python library was used to establish a connection between the Python application and the MySQL database.
4. **PyInstaller**
    - A tool used to convert the Python scripts into a standalone .exe file, allowing the application to run on any Windows system without requiring Python installation.

Flow Diagram



Source Code:

1. db.py

```
import mysql.connector

DB_CONFIG = {
    'host': 'localhost',
    'user': 'root',
    'password': 'admin',
    'database': 'farmit'
}

def get_connection():
    return mysql.connector.connect(**DB_CONFIG)

def fetch_all(query, values=()):
    conn = get_connection()
    cursor = conn.cursor(dictionary=True)
    cursor.execute(query, values)
    result = cursor.fetchall()
    conn.close()
    return result

def execute_query(query, values=()):
```

```
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute(query, values)
    conn.commit()
    conn.close()
```

2. admin.py

```
import tkinter as tk
from tkinter import ttk, messagebox
from db import fetch_all, execute_query
import os
from PIL import Image,ImageTk

def open_admin_dashboard():
    login = tk.Toplevel()
    image = Image.open(r"loginbg.jpg")
    image = image.resize((500,500), Image.Resampling.LANCZOS)
    bg_image = ImageTk.PhotoImage(image)
    bg_label = tk.Label(login, image=bg_image)
    bg_label.image = bg_image
    bg_label.place(x=0, y=0, relwidth=1, relheight=1)
    login.title("Admin Login")
    login.geometry("500x500")

    tk.Label(login, text="Username",font=("Helvetica", 20)).pack(pady=50)
```

```
    username_entry = tk.Entry(login)
    username_entry.pack()

    tk.Label(login, text="Password",font=("Helvetica", 20)).pack(pady=50)
    password_entry = tk.Entry(login, show="*")
    password_entry.pack()

    def authenticate():
        user = username_entry.get()
        pw = password_entry.get()
        result = fetch_all("SELECT * FROM admins WHERE username=%s AND
password=%s", (user, pw))
        if result:
            login.destroy()
            show_admin_dashboard()
        else:
            messagebox.showerror("Login Failed", "Invalid credentials")

    tk.Button(login, text="Login", command=authenticate).pack(pady=30)

def show_admin_dashboard():
    window = tk.Toplevel()
    window.title("Admin Dashboard")
    window.attributes('-fullscreen',True)
```

```
    tk.Button(window, text="Home", command=lambda: [window.destroy(),
os.system("python main.py")], bg="orange", fg="white", font=("Arial",
12)).pack(pady=10)

    style = ttk.Style()
    style.theme_use("default")

# Change Treeview background
    style.configure("Treeview",
    background="#e8f5e9",     # light green
    foreground="black",
    rowheight=25,
    fieldbackground="#e8f5e9")

# Change selected row highlight color
    style.map("Treeview",
    background=[("selected", "#a5d6a7")])

    notebook = ttk.Notebook(window)
    notebook.pack(fill=tk.BOTH, expand=True)

    farmer_tab = ttk.Frame(notebook)
    customer_tab = ttk.Frame(notebook)
    notebook.add(farmer_tab, text="Farmers")
    notebook.add(customer_tab, text="Customers")

    def load_table(tab, query, delete_query, update_query=None):
```

```
for widget in tab.winfo_children():
    widget.destroy()

data = fetch_all(query)
cols = list(data[0].keys()) if data else []

tree = ttk.Treeview(tab, columns=cols, show="headings")
for col in cols:
    tree.heading(col, text=col)
    tree.column(col, width=120)

for row in data:
    tree.insert("", "end", iid=row['id'], values=tuple(row.values()))
tree.pack(fill=tk.BOTH, expand=True)

def delete_selected():
    selected = tree.selection()
    for sel in selected:
        execute_query(delete_query, (int(sel),))
    load_all()

def edit_selected():
    selected = tree.selection()
    if not selected:
        return
    row = tree.item(selected[0])['values']
```

```
        edit_window = tk.Toplevel()
        edit_window.title("Edit Entry")
        entries = []
        for i, col in enumerate(cols):
            tk.Label(edit_window, text=col).pack()
            e = tk.Entry(edit_window)
            e.insert(0, row[i])
            e.pack()
            entries.append(e)

        def save_changes():
            values = [e.get() for e in entries[1:]]  # skip id
            values.append(entries[0].get())  # id last
            execute_query(update_query, tuple(values))
            edit_window.destroy()
            load_all()

        tk.Button(edit_window, text="Save", command=save_changes).pack()

    if update_query:
        tk.Button(tab, text="Edit Selected", command=edit_selected, bg="blue",
fg="white").pack(pady=5)
    tk.Button(tab, text="Delete Selected", command=delete_selected, bg="red",
fg="white").pack(pady=5)

def load_all():
```

```
    load_table(farmer_tab, "SELECT * FROM farmers", "DELETE FROM farmers WHERE id = %s", "UPDATE farmers SET username=%s, password=%s, name=%s, product=%s, contact=%s, tag=%s, location=%s WHERE id=%s")

    load_table(customer_tab, "SELECT * FROM customers", "DELETE FROM customers WHERE id = %s")


load_all()
```

3. customer.py

```
import tkinter as tk
from tkinter import messagebox, ttk
from db import fetch_all, execute_query
import os
from PIL import Image,ImageTk

def open_customer_interface():
    login_window = tk.Toplevel()
    image = Image.open(r"loginbg.jpg")
    image = image.resize((500,500), Image.Resampling.LANCZOS)
    bg_image = ImageTk.PhotoImage(image)
    bg_label = tk.Label(login_window, image=bg_image)
    bg_label.image = bg_image
    bg_label.place(x=0, y=0, relwidth=1, relheight=1)

    login_window.title("Customer Login")
    login_window.geometry("500x500")
```

```
    tk.Label(login_window, text="Username",font=("Helvetica", 20)).pack(pady=50)
    username_entry = tk.Entry(login_window)
    username_entry.pack()

    tk.Label(login_window, text="Password",font=("Helvetica", 20)).pack(pady=50)
    password_entry = tk.Entry(login_window, show="*")
    password_entry.pack()

    def login():
        username = username_entry.get().strip()
        password = password_entry.get().strip()

        result = fetch_all("SELECT * FROM customers WHERE username=%s AND
password=%s", (username, password))
        if result:
            login_window.destroy()
            show_customer_dashboard(result[0])
        else:
            messagebox.showerror("Login Failed", "Invalid credentials")

    tk.Button(login_window, text="Login", command=login).pack(pady=30)

def show_customer_dashboard(customer):
    window = tk.Toplevel()
    window.title(f"Customer Dashboard - {customer['name']}")
```

```
    window.attributes('-fullscreen',True)

    tk.Button(window, text="Home", command=lambda: [window.destroy(),
os.system("python main.py")], bg="orange", fg="white", font=("Arial",
12)).pack(pady=10)

    style = ttk.Style()
    style.theme_use("default")

# Change Treeview background
    style.configure("Treeview",
    background="#e8f5e9",     # light green
    foreground="black",
    rowheight=25,
    fieldbackground="#e8f5e9")

# Change selected row highlight color
    style.map("Treeview",
    background=[("selected", "#a5d6a7")])

    notebook = ttk.Notebook(window)
    notebook.pack(fill=tk.BOTH, expand=True)

    browse_tab = ttk.Frame(notebook)
    wishlist_tab = ttk.Frame(notebook)
    notebook.add(browse_tab, text="Browse Products")
    notebook.add(wishlist_tab, text="My Wishlist")
```

```
search_frame = tk.Frame(browse_tab)
search_frame.pack(pady=10)

tk.Label(search_frame, text="Search: ").grid(row=0, column=0, padx=5)
search_entry = tk.Entry(search_frame)
search_entry.grid(row=0, column=1, padx=5)

result_frame = tk.Frame(browse_tab)
result_frame.pack(fill=tk.BOTH, expand=True)

columns = ["ID", "Username", "Name", "Product", "Contact", "Tag", "Location"]
tree = ttk.Treeview(result_frame, columns=columns, show="headings")
for col in columns:
    tree.heading(col, text=col)
    tree.column(col, width=120)
tree.pack(fill=tk.BOTH, expand=True)

def display_results(results):
    for row in tree.get_children():
        tree.delete(row)
    for f in results:
        tree.insert("", "end", values=(
            f.get("id"), f.get("username"), f.get("name"), f.get("product"),
            f.get("contact"), f.get("tag"), f.get("location")
        ))
```

```
    def search():
        q = search_entry.get().strip().lower()
        results = fetch_all("""
            SELECT * FROM farmers
            WHERE LOWER(tag) LIKE %s OR LOWER(location) LIKE %s OR
LOWER(product) LIKE %s
        """, (f"%{q}%", f"%{q}%", f"%{q}%"))
        display_results(results)


    def view_selected():
        selected = tree.selection()
        if not selected:
            tk.Label(info_window, text= "Please select a farmer", font=("Arial",
12)).pack(pady=5)
            return


        row = tree.item(selected[0])['values']
        info_window = tk.Toplevel(window)
        info_window.title("Farmer Info")
        info_window.geometry("400x300")


        fields = tree["columns"]
        for i, field in enumerate(fields):
            tk.Label(info_window, text=f"{field.capitalize()}: {row[i]}", font=("Arial",
12)).pack(pady=5)
```

```
    def save_to_wishlist():
        selected = tree.selection()
        if not selected:
            tk.Label(info_window, text="Please select a farmer", font=("Arial",
12)).pack(pady=5)
            return

        row = tree.item(selected[0])['values']
        info_window = tk.Toplevel(window)
        info_window.title("WishList Info")
        info_window.geometry("350x100")

        farmer_id = row[0]
        customer_id = customer['id']
        existing = fetch_all("SELECT * FROM wishlist WHERE customer_id=%s AND
farmer_id=%s", (customer_id, farmer_id))
        if existing:
            tk.Label(info_window, text=f"{row[2]}'s product is already in your wishlist.",
font=("Arial", 12)).pack(pady=5)
        else:
            execute_query("INSERT INTO wishlist (customer_id, farmer_id) VALUES
(%s, %s)", (customer_id, farmer_id))
            tk.Label(info_window, text= f"{row[2]}'s product is added to your wishlist.",
font=("Arial", 12)).pack(pady=5)
            load_wishlist()

    action_frame = tk.Frame(browse_tab)
    action_frame.pack(pady=5)
```

```
    tk.Button(action_frame, text="View Selected", command=view_selected, bg="blue", fg="white").pack(side=tk.LEFT, padx=5)
    tk.Button(action_frame, text="Save to Wishlist", command=save_to_wishlist, bg="green", fg="white").pack(side=tk.LEFT, padx=5)

    tk.Button(search_frame, text="Search", command=search).grid(row=0, column=2, padx=5)
    display_results(fetch_all("SELECT * FROM farmers"))

    wishlist_tree = ttk.Treeview(wishlist_tab, columns=columns, show="headings")
    for col in columns:
        wishlist_tree.heading(col, text=col)
        wishlist_tree.column(col, width=120)
    wishlist_tree.pack(fill=tk.BOTH, expand=True, padx=10, pady=10)

    def load_wishlist():
        for row in wishlist_tree.get_children():
            wishlist_tree.delete(row)
        results = fetch_all("""
            SELECT f.* FROM farmers f
            JOIN wishlist w ON f.id = w.farmer_id
            WHERE w.customer_id = %s
        """, (customer['id'],))
        for f in results:
            wishlist_tree.insert("", "end", values=(
                f.get("id"), f.get("username"), f.get("name"), f.get("product"),
                f.get("contact"), f.get("tag"), f.get("location")
```

```
        ))

    load_wishlist()

    def remove_from_wishlist():
        selected = wishlist_tree.selection()
        if not selected:
            tk.Label(info_window, text= "Please select a farmer", font=("Arial",
12)).pack(pady=5)
            return
        row = wishlist_tree.item(selected[0])['values']
        info_window = tk.Toplevel(window)
        info_window.title("Wishlist info")
        info_window.geometry("350x100")
        farmer_id = row[0]
        customer_id = customer['id']

        execute_query("DELETE FROM wishlist WHERE customer_id=%s AND
farmer_id=%s", (customer_id, farmer_id))
        tk.Label(info_window, text= f"{row[2]}'s product removed from wishlist.",
font=("Arial", 12)).pack(pady=5)
        load_wishlist()

    tk.Button(wishlist_tab, text="Remove from Wishlist",
command=remove_from_wishlist, bg="red", fg="white").pack(pady=5)
```

4.farmer.py

```
import tkinter as tk
from tkinter import messagebox
from db import fetch_all, execute_query
import os
from PIL import Image,ImageTk

def open_farmer_interface():
    login_window = tk.Toplevel()
    image = Image.open(r"loginbg.jpg")
    image = image.resize((500,500), Image.Resampling.LANCZOS)
    bg_image = ImageTk.PhotoImage(image)
    bg_label = tk.Label(login_window, image=bg_image)
    bg_label.image = bg_image
    bg_label.place(x=0, y=0, relwidth=1, relheight=1)

    login_window.title("Farmer Login")
    login_window.geometry("500x500")

    tk.Label(login_window, text="Username",font=("Helvetica", 20)).pack(pady=50)
    username_entry = tk.Entry(login_window)
    username_entry.pack()

    tk.Label(login_window, text="Password",font=("Helvetica", 20)).pack(pady=50)
    password_entry = tk.Entry(login_window, show="*")
```

```
    password_entry.pack()

    def login():
        username = username_entry.get().strip()
        password = password_entry.get().strip()

        result = fetch_all("SELECT * FROM farmers WHERE username=%s AND
password=%s", (username, password))
        if result:
            login_window.destroy()
            show_farmer_info(result[0])
        else:
            messagebox.showerror("Error", "Invalid credentials")

    tk.Button(login_window, text="Login", command=login).pack(pady=30)

def show_farmer_info(farmer):
    window = tk.Toplevel()
    window.title("Farmer Dashboard")
    window.attributes('-fullscreen',True)

    image = Image.open(r"farmerbg.jpg")
    image = image.resize((1600,900), Image.Resampling.LANCZOS)
    bg_image = ImageTk.PhotoImage(image)
    bg_label = tk.Label(window, image=bg_image)
    bg_label.image = bg_image
```

```
    bg_label.place(x=0, y=0, relwidth=1, relheight=1)

    tk.Button(window, text="Home", command=lambda: [window.destroy(),
os.system("python main.py")], bg="orange", fg="white", font=("Arial",
12)).pack(pady=10)



    fields = ["product", "contact", "tag", "location"]
    entries = {}

    for field in fields:
        tk.Label(window, text=f"{field.capitalize()}",font=("Helvetica",
20)).pack(pady=20)
        entry = tk.Entry(window, width=30)
        entry.insert(0, farmer.get(field, ""))
        entry.pack()
        entries[field] = entry

    def update_info():
        info_window = tk.Toplevel(window)
        info_window.title("Wishlist info")
        info_window.geometry("350x100")
        updates = {f: entries[f].get().strip() for f in fields}
        query = "UPDATE farmers SET product=%s, contact=%s, tag=%s, location=%s
WHERE id=%s"
        execute_query(query, (updates["product"], updates["contact"], updates["tag"],
updates["location"], farmer["id"]))
```

```
        tk.Label(info_window, text= "Information updated.", font=("Arial",
12)).pack(pady=5)


    tk.Button(window, text="Update Info", command=update_info,font=("Helvetica",
20),bg="#4CAF50", fg="white").pack(pady=30)
```

5. register.py

```
import tkinter as tk
from tkinter import messagebox
from db import execute_query,fetch_all


def open_registration_window():
    window = tk.Toplevel()
    window.title("Register")
    window.geometry("600x700")


    tk.Label(window, text="Register As", font=("Arial", 20, "bold")).pack(pady=30)
    role_var = tk.StringVar(value="farmer")
    tk.Radiobutton(window, text="Farmer", variable=role_var,
value="farmer").pack(pady=10)
    tk.Radiobutton(window, text="Customer", variable=role_var,
value="customer").pack(pady=10)


    entries = {}
    fields = ["Username", "Password", "Name"]
    for field in fields:
```

```
    tk.Label(window, text=field).pack()
    entry = tk.Entry(window)
    entry.pack()
    entries[field.lower()] = entry

additional = {
    "farmer": ["Product (with price)", "Contact", "Tag", "Location"],
    "customer": []
}
additional_entries = {}
dynamic_frame = tk.Frame(window)
dynamic_frame.pack(pady=30)

def update_form():
    for widget in dynamic_frame.winfo_children():
        widget.destroy()
    additional_entries.clear()
    for field in additional[role_var.get()]:
        tk.Label(dynamic_frame, text=field).pack()
        entry = tk.Entry(dynamic_frame)
        entry.pack()
        additional_entries[field.lower()] = entry

role_var.trace("w", lambda *args: update_form())
update_form()
```

```
    def register():
        username = entries["username"].get().strip()
        password = entries["password"].get().strip()
        name = entries["name"].get().strip()
        role = role_var.get()

        if not (username and password and name):
            messagebox.showerror("Error", "Please fill all basic fields")
            return

        if role == "farmer":
            product = additional_entries["product (with price)"].get().strip()
            contact = additional_entries["contact"].get().strip()
            tag = additional_entries["tag"].get().strip()
            location = additional_entries["location"].get().strip()

            if not all([product, contact, tag, location]):
                    messagebox.showerror("Error", "Please fill all farmer fields")

            if not username or not password or not name:
                messagebox.showwarning("Input Error", "All fields are required.")
                return

            existing = fetch_all("SELECT * FROM farmers WHERE username=%s",
(username,))
            if existing:
```

```
            messagebox.showerror("Registration Failed", "Username already exists!")
            return

        query = "INSERT INTO farmers (username, password, name, product, contact,
tag, location) VALUES (%s, %s, %s, %s, %s, %s, %s)"
        values = (username, password, name, product, contact, tag, location)
    else:
        existing = fetch_all("SELECT * FROM customers WHERE username=%s",
(username,))
        if existing:
            messagebox.showerror("Registration Failed", "Username already exists!")
            return

        query = "INSERT INTO customers (username, password, name) VALUES
(%s, %s, %s)"
        values = (username, password, name)

    try:
        execute_query(query, values)
        messagebox.showinfo("Success", f"{role.capitalize()} registered successfully")
        window.destroy()
    except Exception as e:
        messagebox.showerror("Database Error", str(e))

tk.Button(window, text="Register", bg="#4CAF50", fg="white",
command=register).pack(pady=30)
```

6. main.py

```
import tkinter as tk
from admin import open_admin_dashboard
from customer import open_customer_interface
from farmer import open_farmer_interface
from register import open_registration_window
from PIL import Image,ImageTk
import os,sys

def main():
    root = tk.Tk()
    image = Image.open(r"bg.jpeg")
    image = image.resize((1550,1020), Image.Resampling.LANCZOS)
    bg_image = ImageTk.PhotoImage(image)
    bg_label = tk.Label(root, image=bg_image)
    bg_label.image = bg_image
    bg_label.place(x=0, y=0, relwidth=1, relheight=1)
    root.title("Farm-It | Login")
    root.attributes('-fullscreen',True)
    root.configure(bg="#f0fff0")

    tk.Label(root, text="Welcome to Farm-It", font=("Helvetica", 50, "bold"),
bg="#228B22",fg="#FFFFFF").pack(pady=90)

    tk.Button(root, text="Admin Login", width=25, font=("Helvetica", 20),
bg="#4CAF50", fg="white", command=open_admin_dashboard).pack(pady=30)
```

```
    tk.Button(root, text="Customer Login", width=25, font=("Helvetica", 20), bg="#2196F3", fg="white", command=open_customer_interface).pack(pady=30)
    tk.Button(root, text="Farmer Login", width=25, font=("Helvetica", 20), bg="#FF9800", fg="white", command=open_farmer_interface).pack(pady=30)
    tk.Button(root, text="Register", width=25, font=("Helvetica", 20), bg="#9C27B0", fg="white", command=open_registration_window).pack(pady=30)
    tk.Button(root, text="Exit", width=25, font=("Helvetica", 20),bg="#f44336", fg="white", command=root.destroy).pack(pady=5)

    root.mainloop()


main()
```

7. SQL Commands

```
CREATE DATABASE IF NOT EXISTS farmit;

USE farmit;

CREATE TABLE admins (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50),
    password VARCHAR(50)
);

CREATE TABLE IF NOT EXISTS farmers (
```

```
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL,
    password VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,
    product VARCHAR(100),         -- nullable
    contact VARCHAR(20),
    tag VARCHAR(50),
    location VARCHAR(100)
);

CREATE TABLE customers (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50),
    password VARCHAR(50),
    name VARCHAR(100)
);

INSERT INTO admins (username, password) VALUES ('admin', 'admin123');

INSERT INTO farmers (username, password, name, product, contact, tag, location)
VALUES
('f1', 'pass1', 'Raj Patel', 'Wheat 100/kg', '9876543210', '#grains', 'Ahmedabad'),
('f2', 'pass2', 'Lakshmi Reddy', 'Rice 90/kg', '9123456789', '#cereals', 'Hyderabad'),
('f3', 'pass3', 'John Singh', 'Mango 60/kg', '9988776655', '#fruits', 'Nagpur'),
('f4', 'pass4', 'Meena Sharma', 'Tomato 30/kg', '9871234567', '#vegetables', 'Jaipur'),
```

('f5', 'pass5', 'Kiran Das', NULL, NULL, NULL, NULL); -- Farmer registered but hasn’t added details yet

```
INSERT INTO customers (username, password, name)
VALUES
('c1', 'cust1', 'Amit Shah'),
('c2', 'cust2', 'Sneha Iyer'),
('c3', 'cust3', 'Rahul Dev'),
('c4', 'cust4', 'Priya Mehta'),
('c5', 'cust5', 'Naveen Kumar');
```

```
CREATE TABLE IF NOT EXISTS wishlist (
    id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT,
    farmer_id INT,
    FOREIGN KEY (customer_id) REFERENCES customers(id),
    FOREIGN KEY (farmer_id) REFERENCES farmers(id)
);
```

```
ALTER TABLE customers ADD UNIQUE (username);
ALTER TABLE farmers ADD UNIQUE (username);
```

8. CMD prompt for .exe creation

```
cd C:\farm_it
```

```
pip install pyinstaller


pyinstaller --onefile --windowed ^
--add-data "bg.jpeg;." ^
--add-data "farmerbg.jpg;." ^
--add-data "loginbg.jpg;." ^
--add-data "admin.py;." ^
--add-data "customer.py;." ^
--add-data "farmer.py;." ^
--add-data "register.py;." ^
--add-data "db.py;." ^
main.py
```

Application Flow:

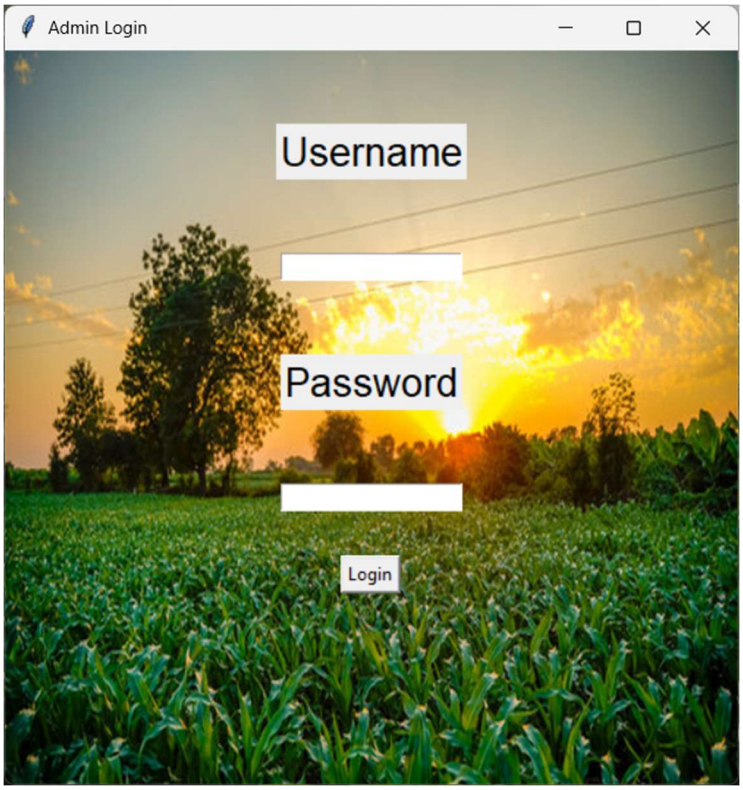
1. **User Selection**
    - The application begins with a home screen offering three login options: **Farmer Login**, **Customer Login**, and **Admin Login**.
2. **Farmer Login**
    - Farmers log in using their credentials.
    - After logging in, they can add, edit, or update their product details (name, price, tag, contact, and location).
    - The updated data is stored in the MySQL database.
3. **Customer Login**
    - Customers log in to browse products.
    - They can search and filter products based on **tags, price range, product name, and location**.
    - Customers can add products to their wishlist for easy access later.
4. **Admin Login**
    - Admins can view all registered farmers and customers in a tabular format.
    - Admins have the authority to delete or modify any farmer profile or product listing to maintain system integrity.
5. **Database Interaction**
    - Every action (login, product search, wishlist management, etc.) communicates with the MySQL database to fetch or store information securely.
6. **Exit and Restart**
    - The user can return to the home screen or exit the application using dedicated buttons.

This flow ensures smooth navigation for all types of users while maintaining data integrity and providing a seamless user experience.

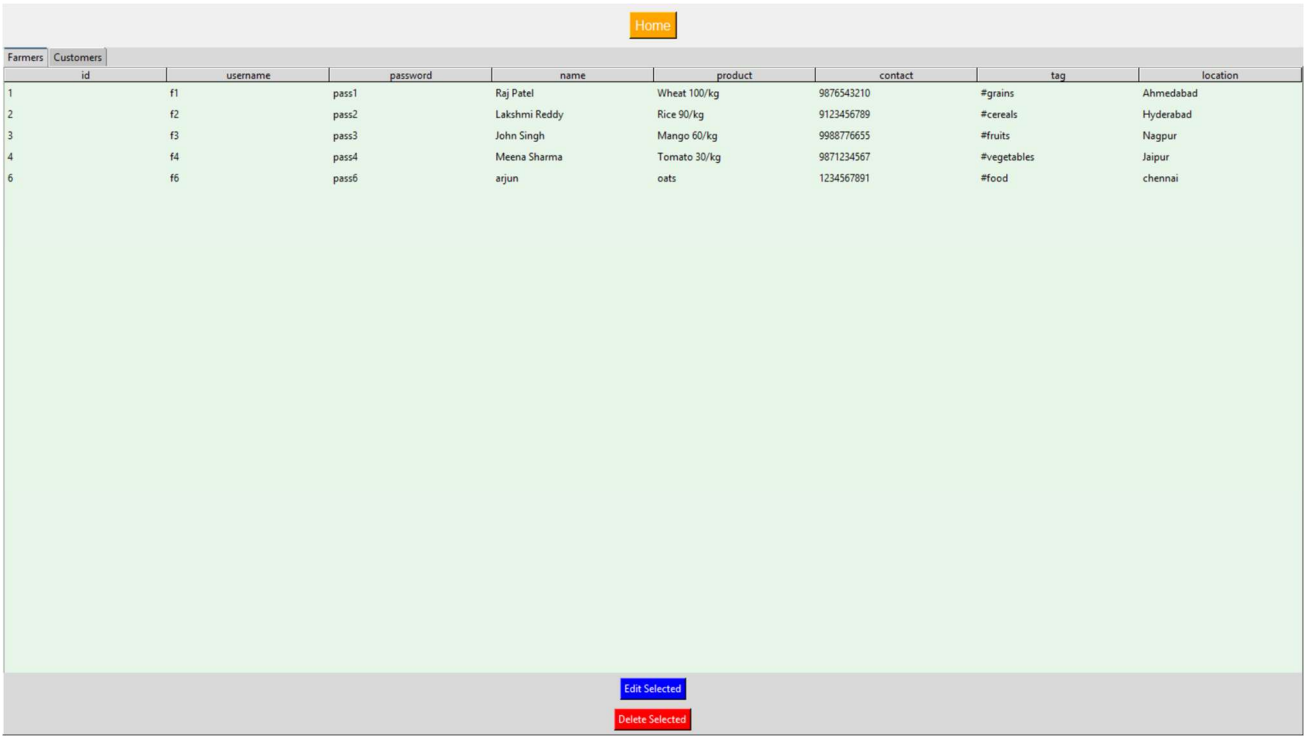
Home Screens:

1. main window

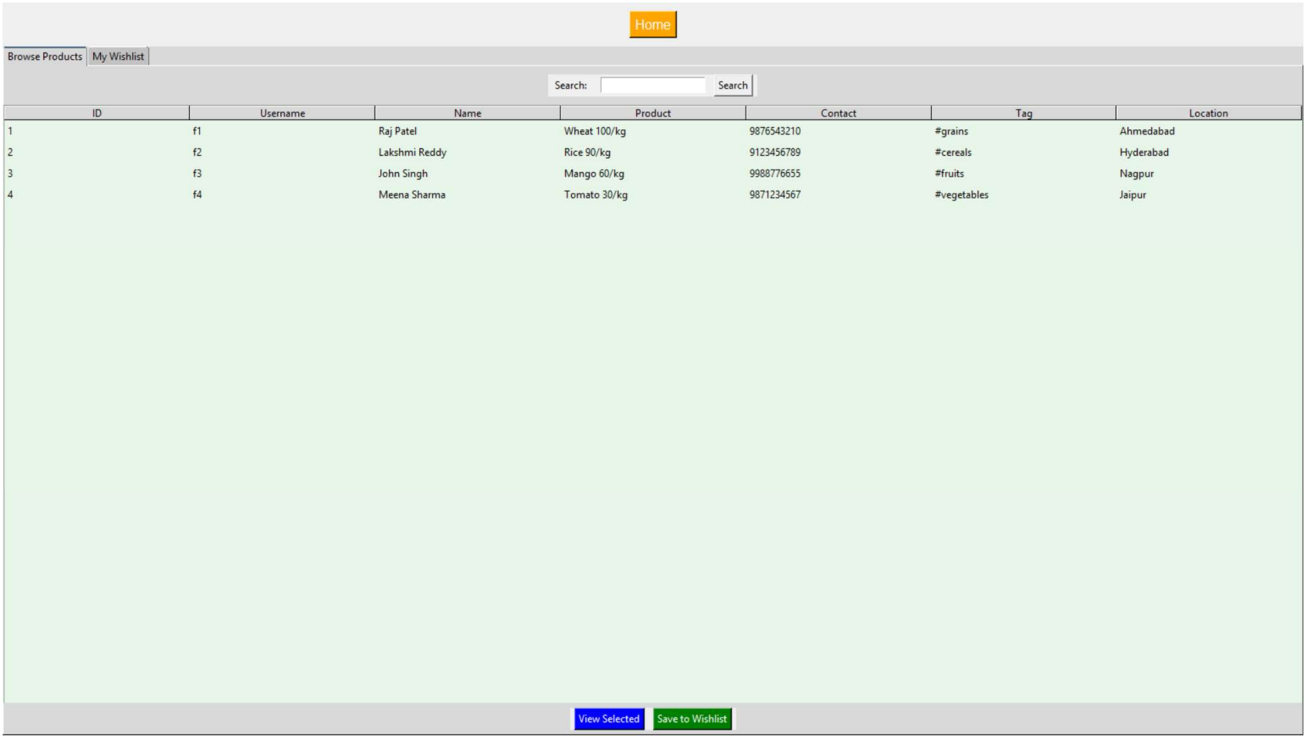
2. login window



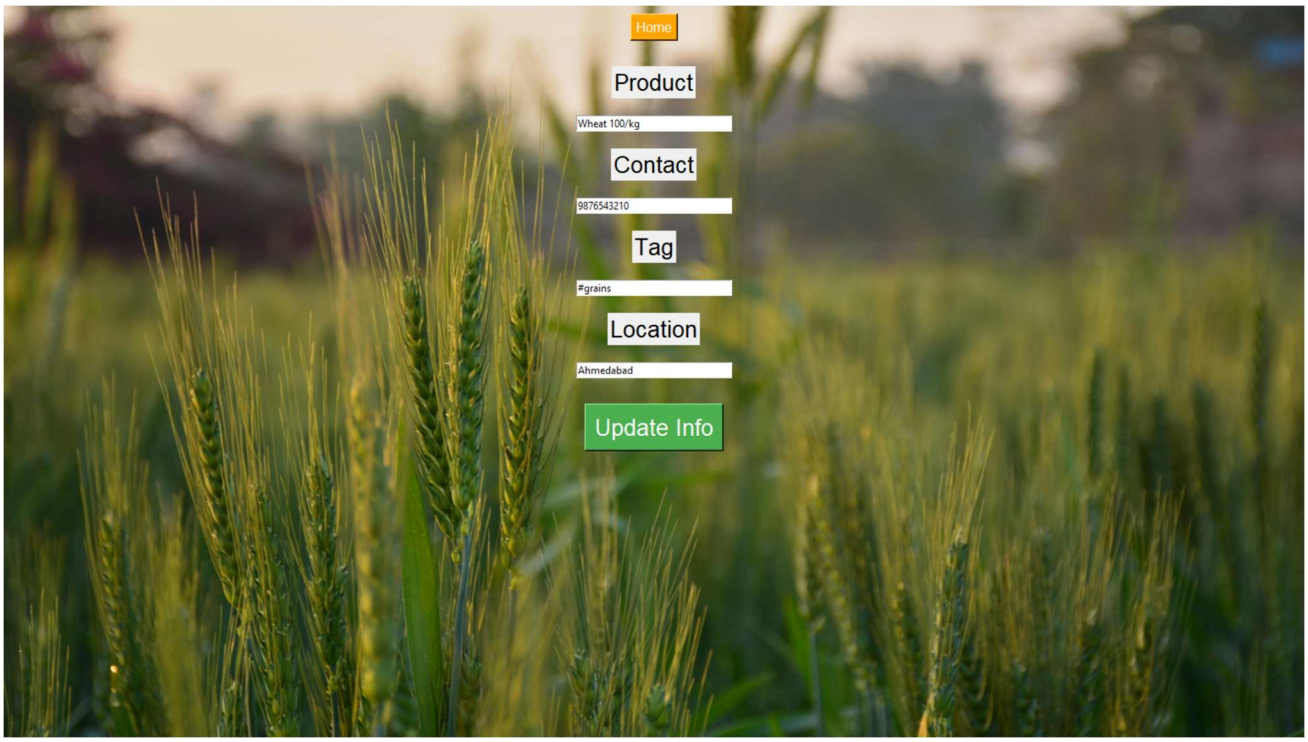
3. Admin dashboard



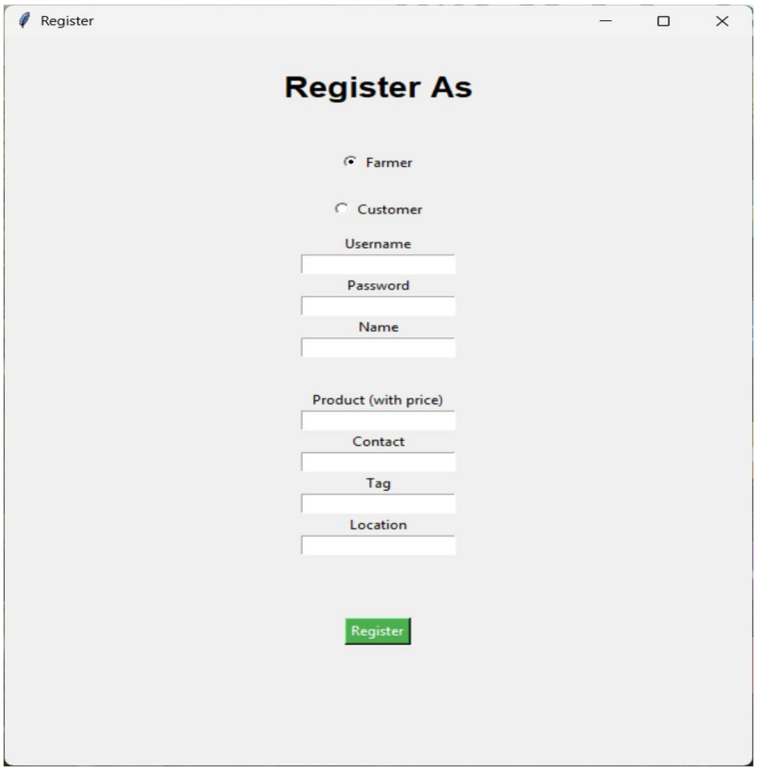
4. Customer dashboard

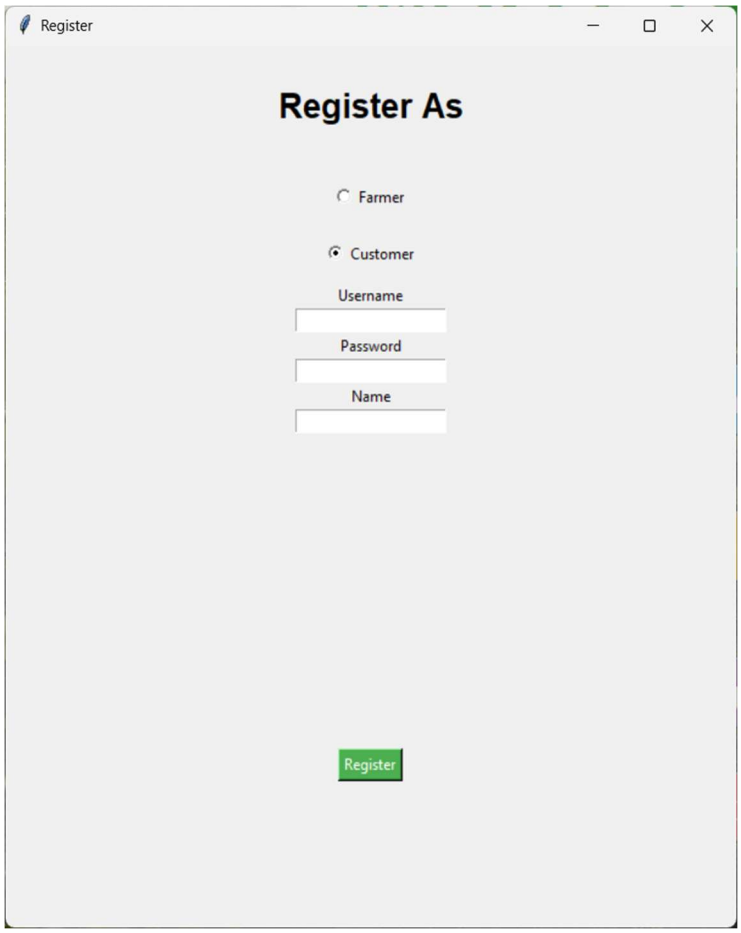


5.Farmer dashboard



6.Register window

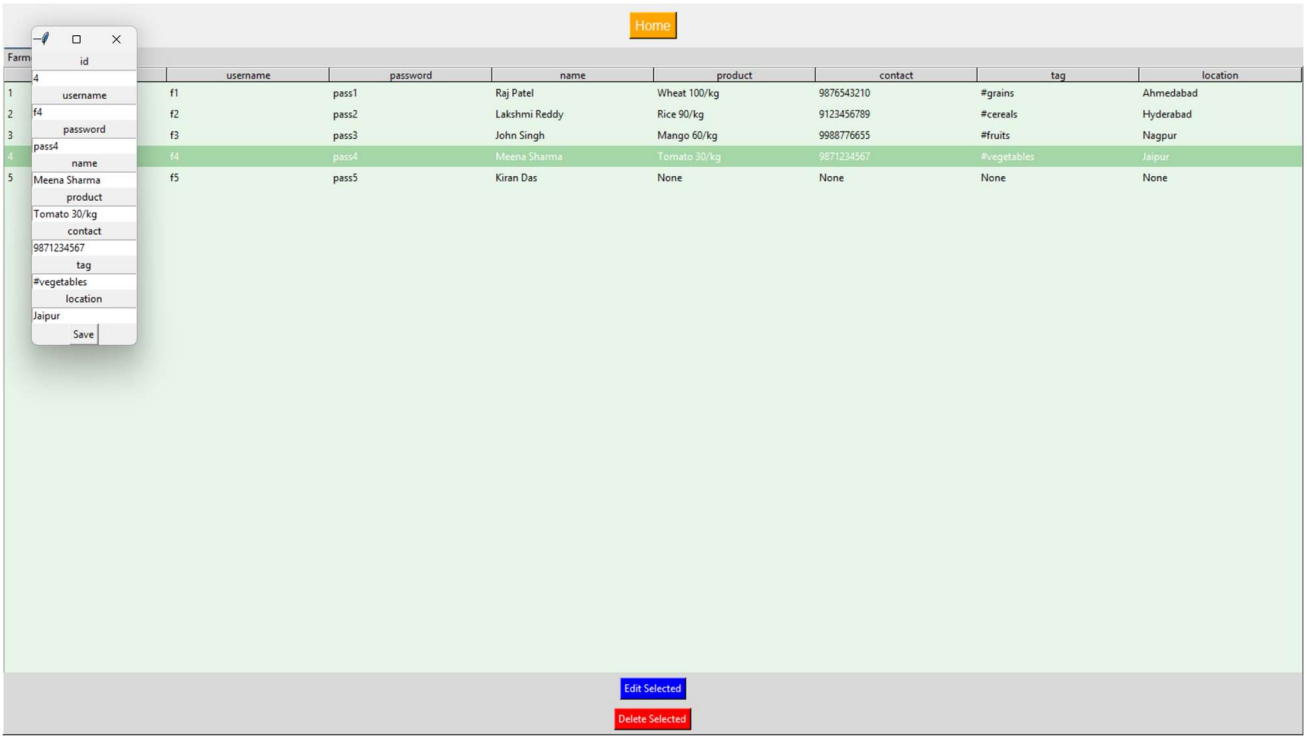



Register
Register As
Farmer
Customer
Username
Password
Name
Register

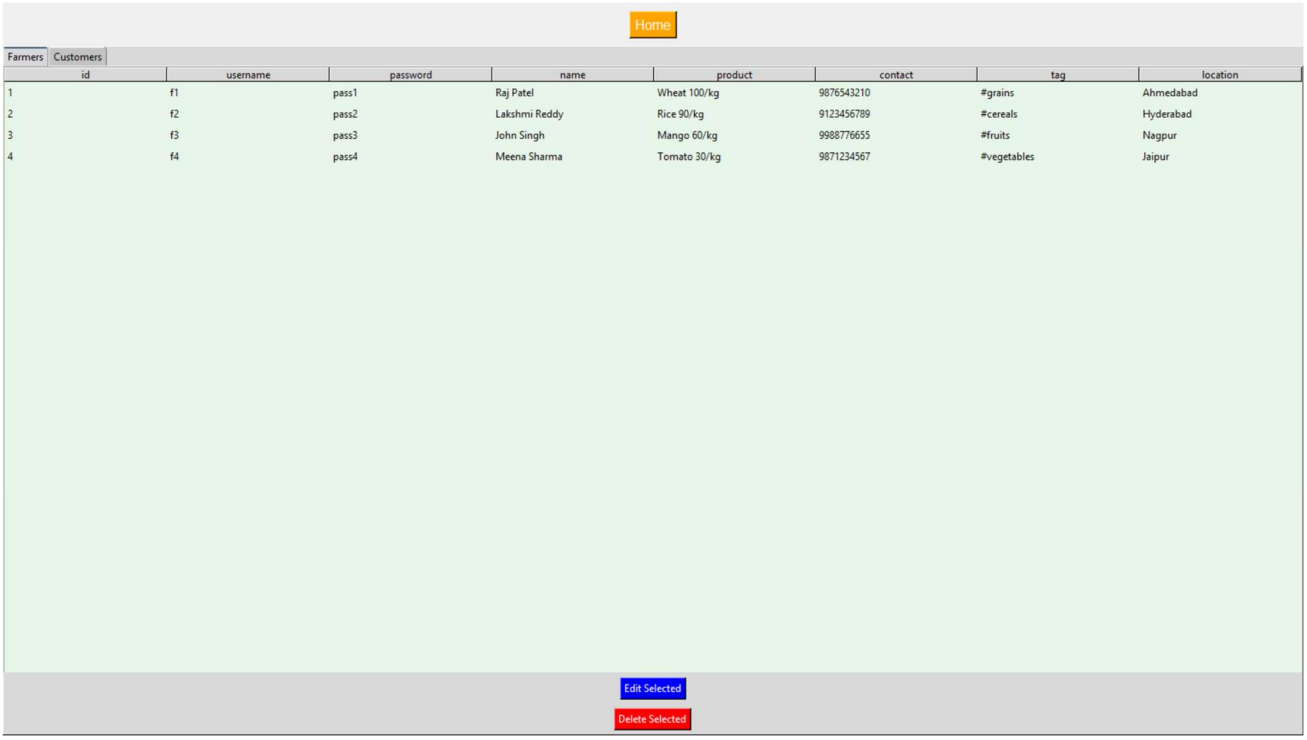
Report Module :

1.admin screen

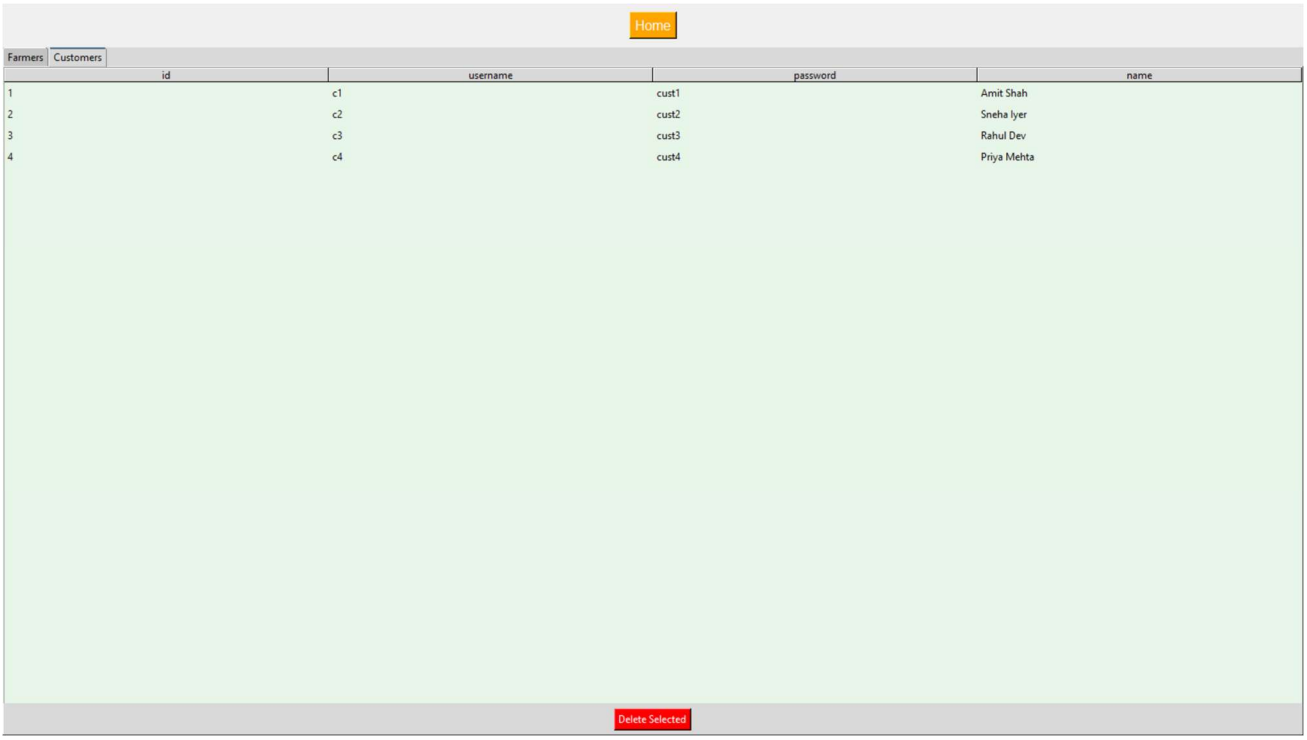
Editing farmer detail

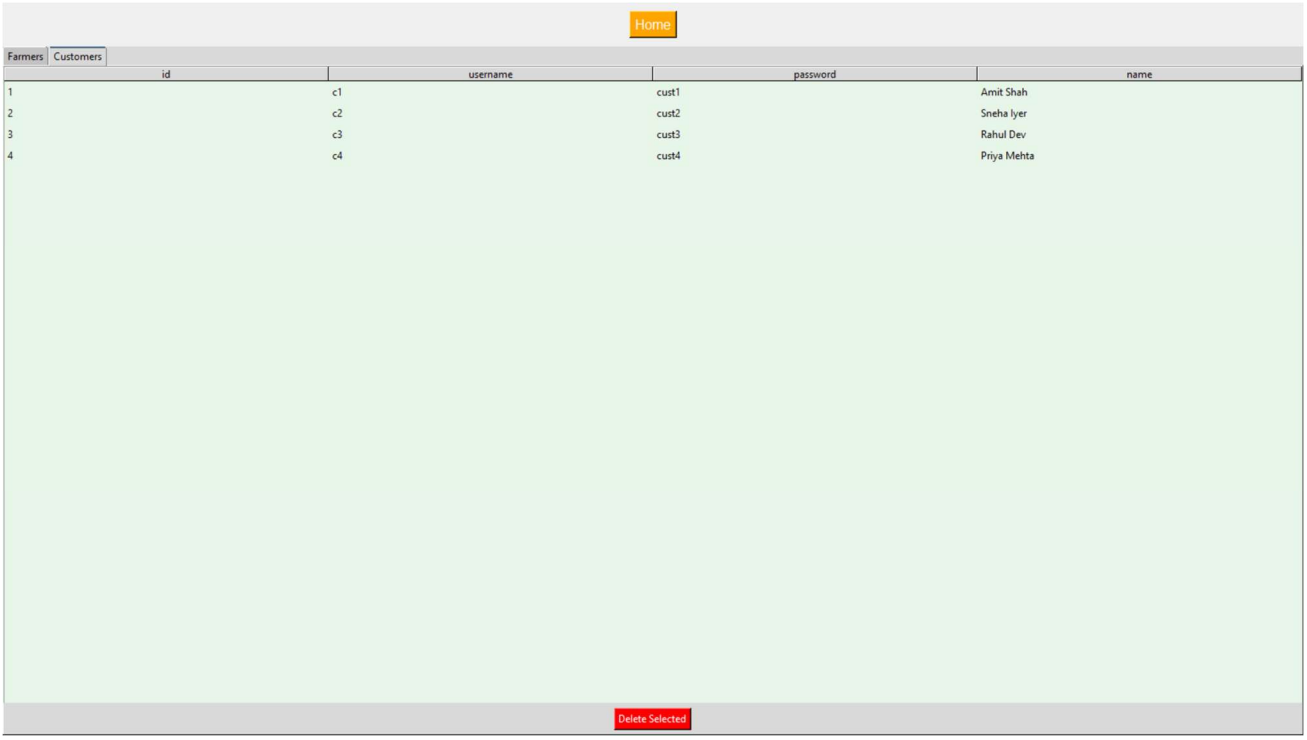


Deleting farmer(here kiran)



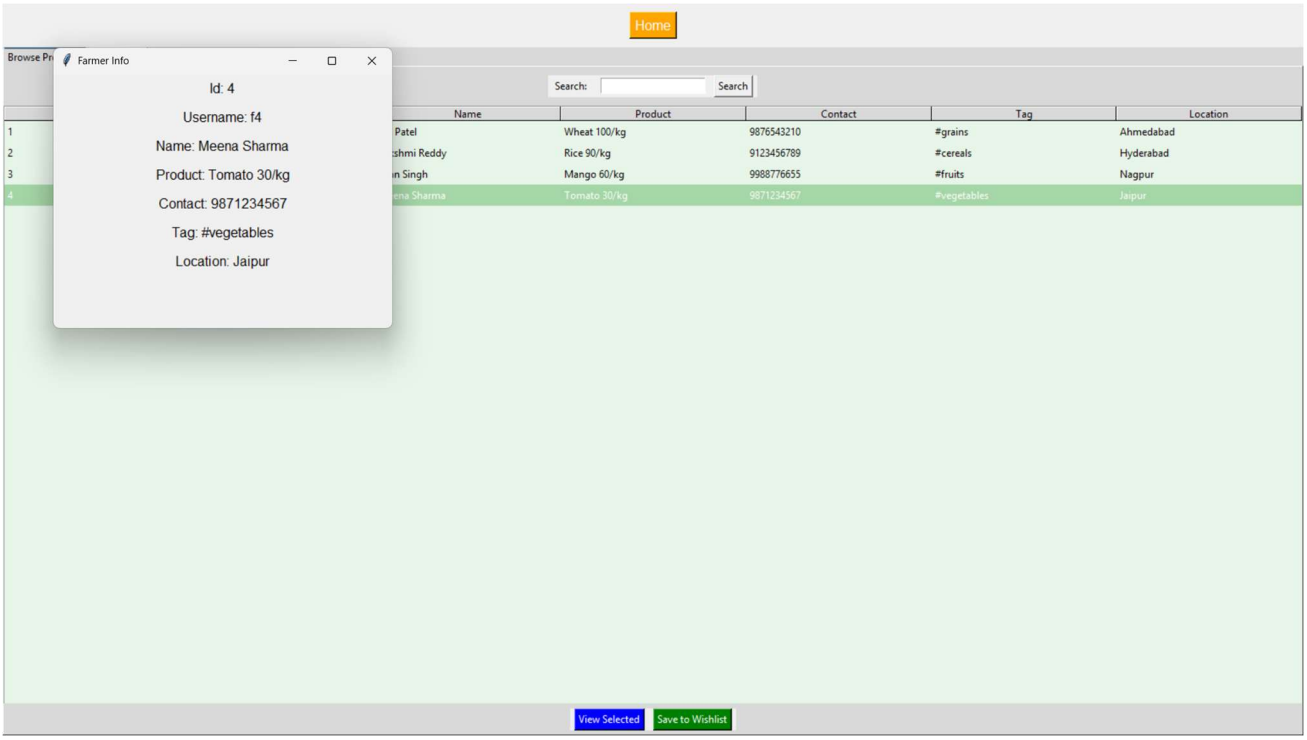
Deleting customer(here c5)



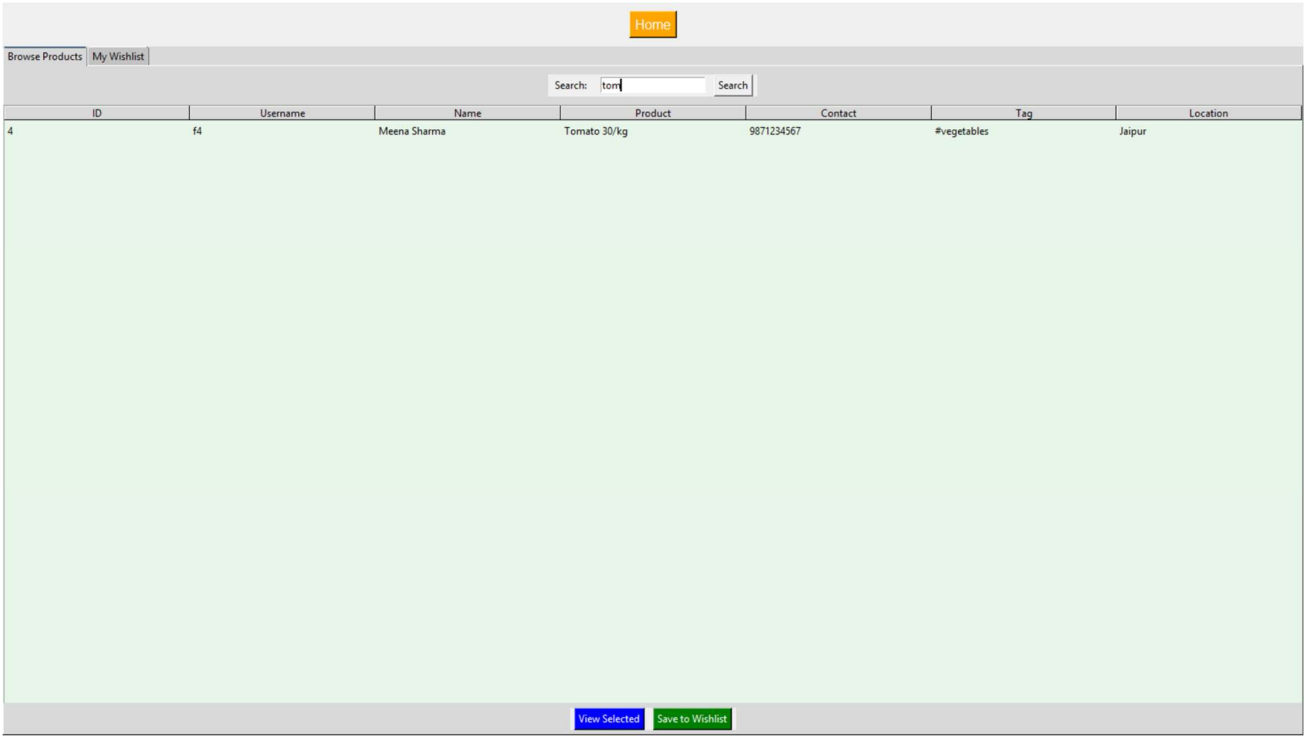


2.customer screen

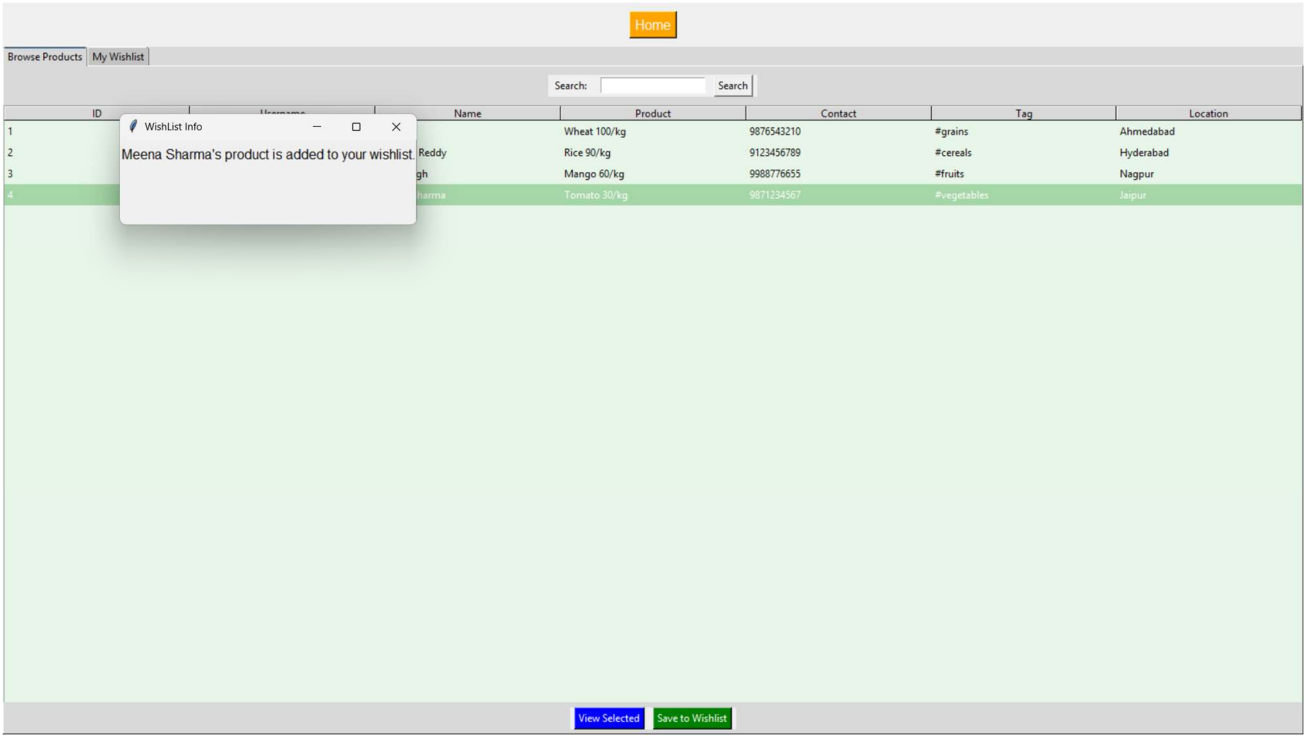
Viewing farmer detail

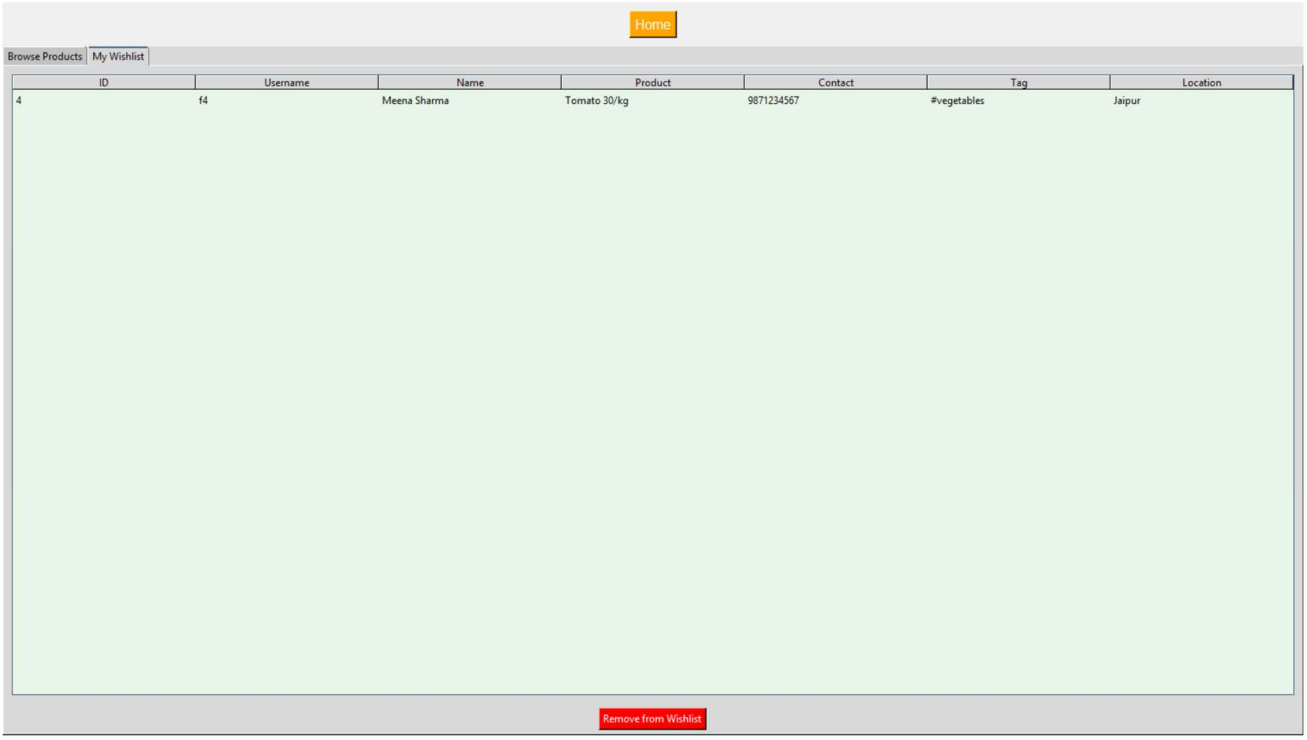


search



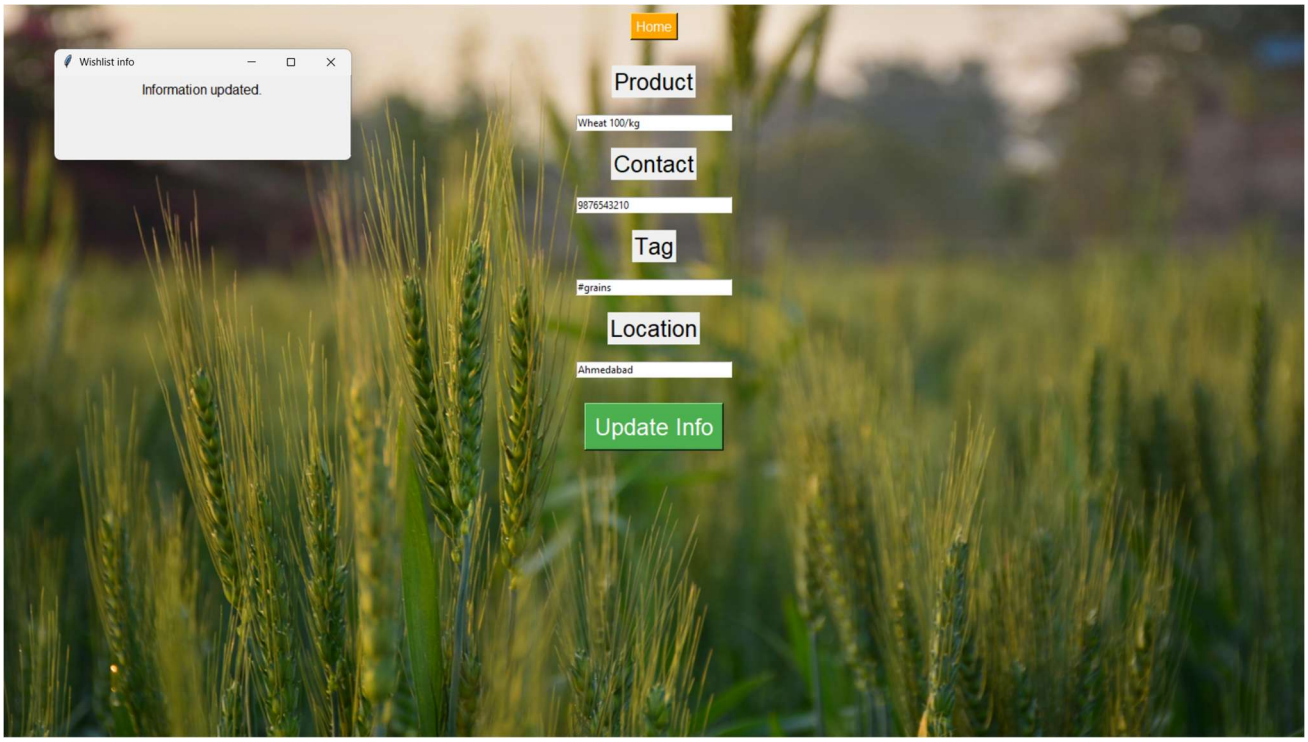
Saving farmer to wishlist





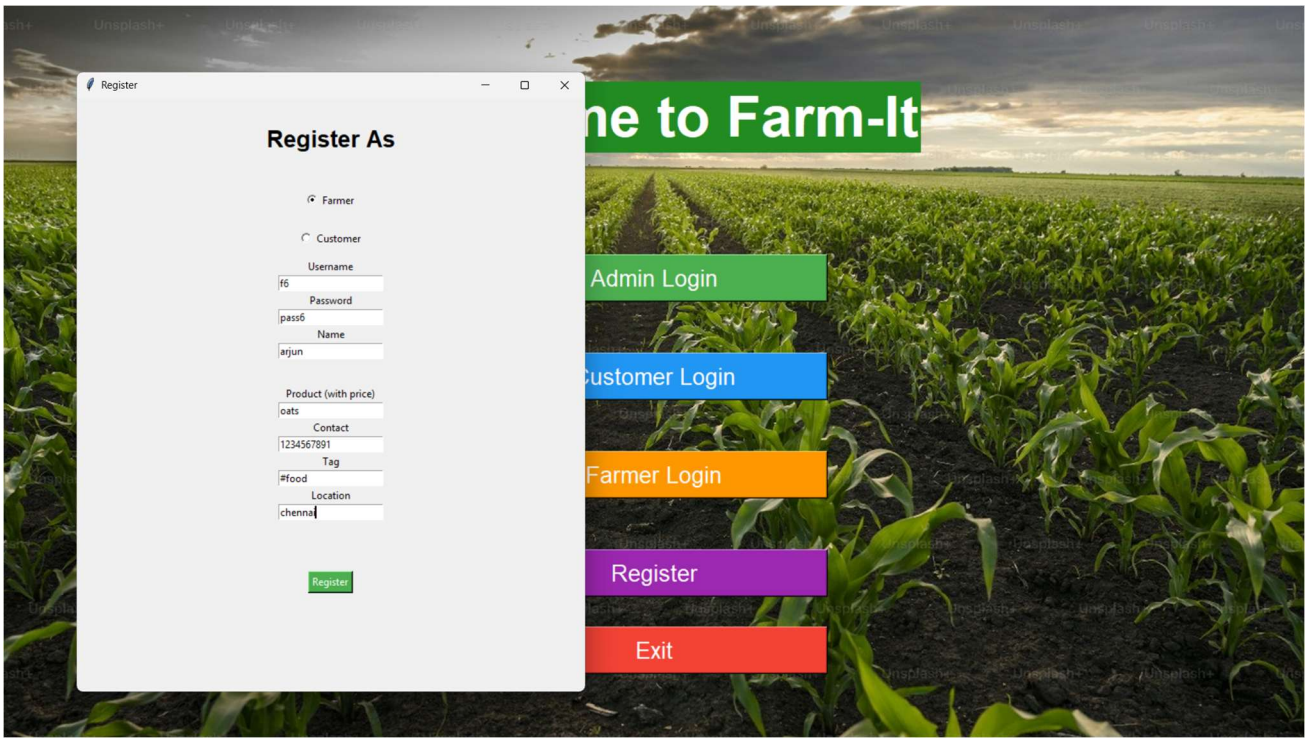
3.farmer screen

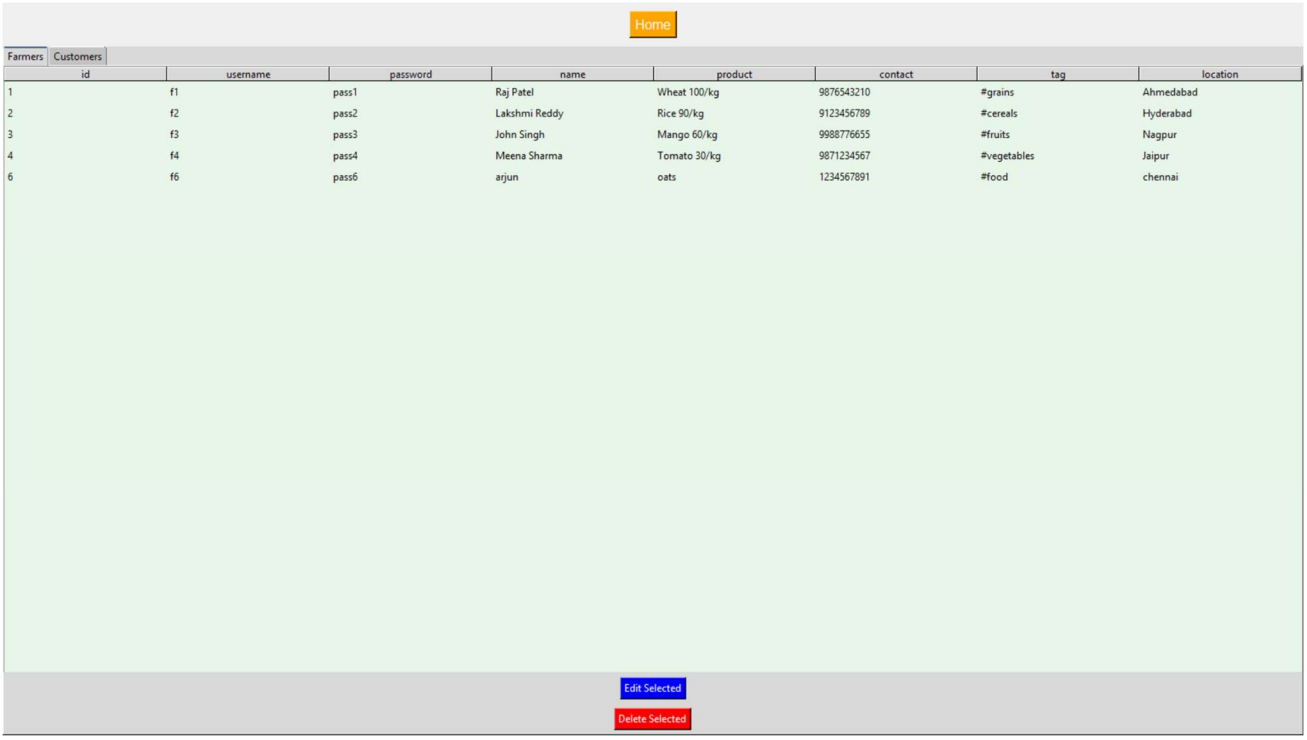
Updating farmer information



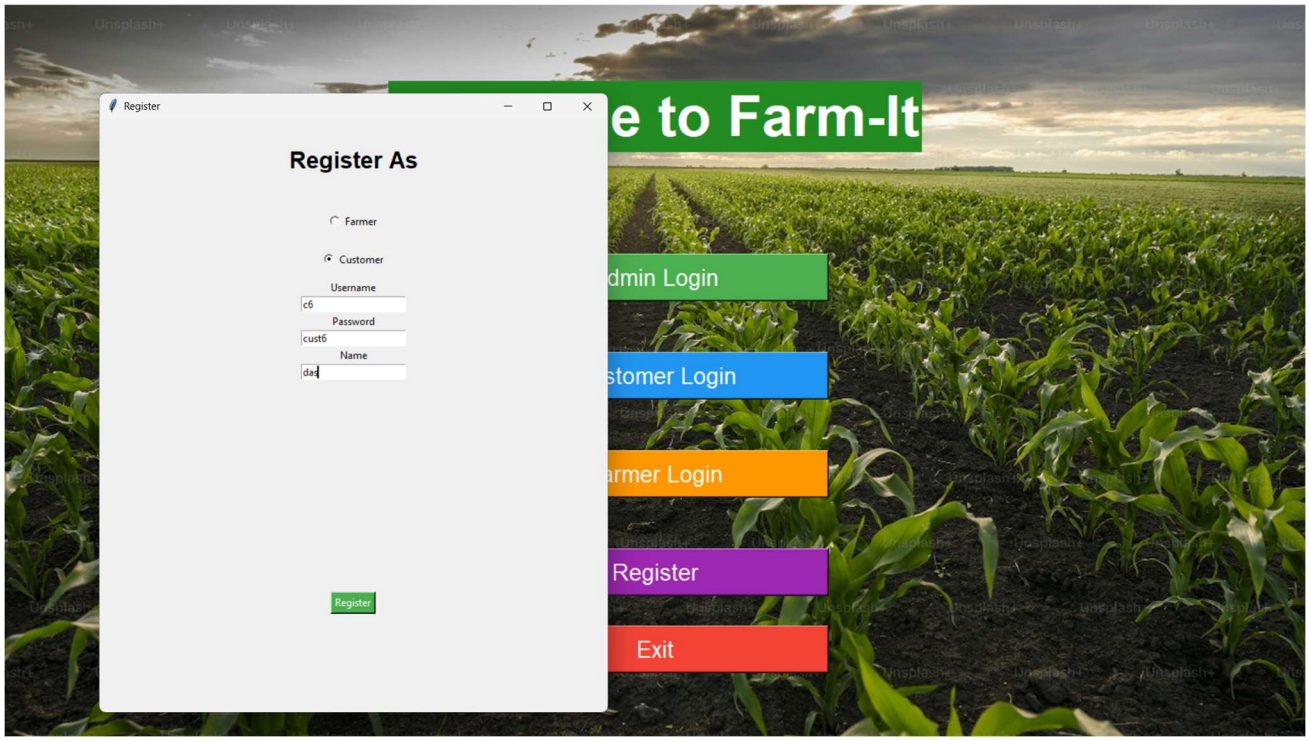
4.Register screen

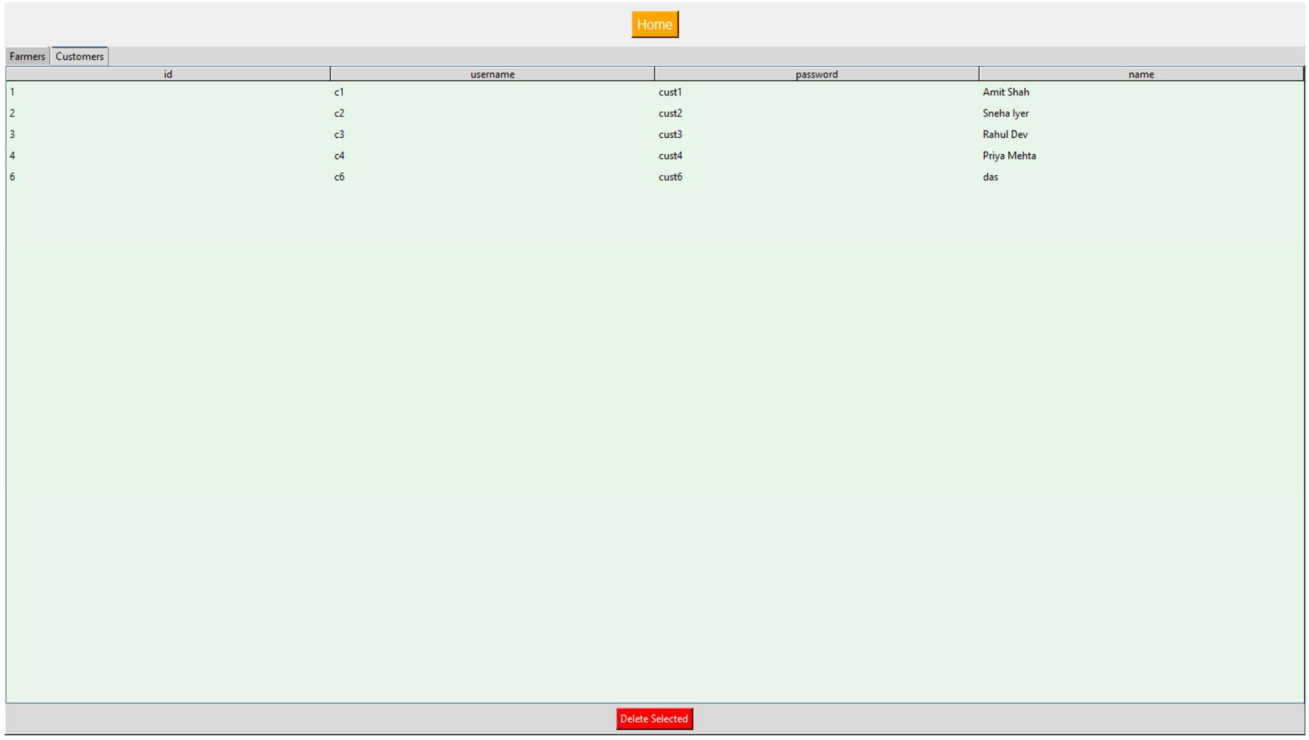
Registering as farmer





Registrering as customer





File part of this project :

Python files:

 db.py

 admin.py

 customer.py

 farmer.py

 register.py

 main.py

Exe File:

 farm_it.exe

Bibliography:

1. https://www.chatgpt.com/ – reference for base code

2. https://www.pinterest.com/ – bg image

3. https://docs.python.org/3/library/tkinter.html - reference for use of tkinter module